

Assignment - 02 Layout

Introduction

In this assignment we are asked to re-enact a given mask in Layout Editor following the given steps and convert the CMOS inverter circuit to Layout using Layout editor of SEMulator3D (Figure 1).

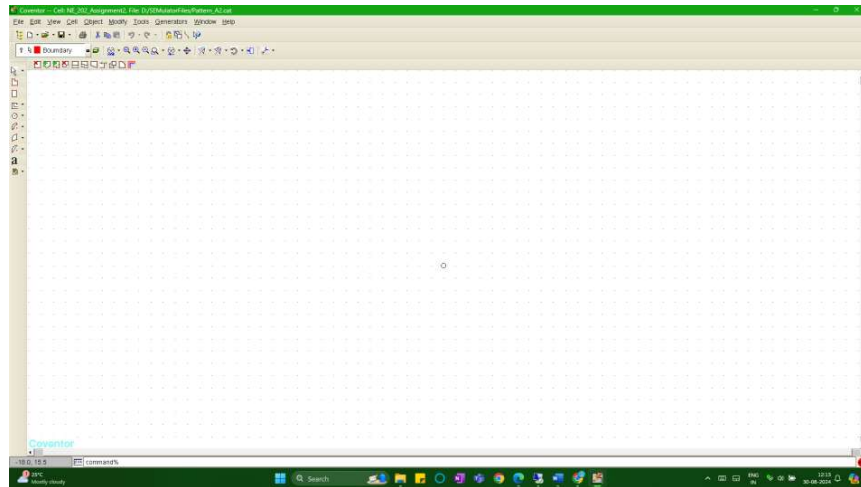


Figure 1: SEMulator3D Layout Editor window

Question 1

For the first task we have been assigned to learn and recreate the given layout. Given steps are as follows -

Step 1: Units are set to nanometres from Tools -> Units.

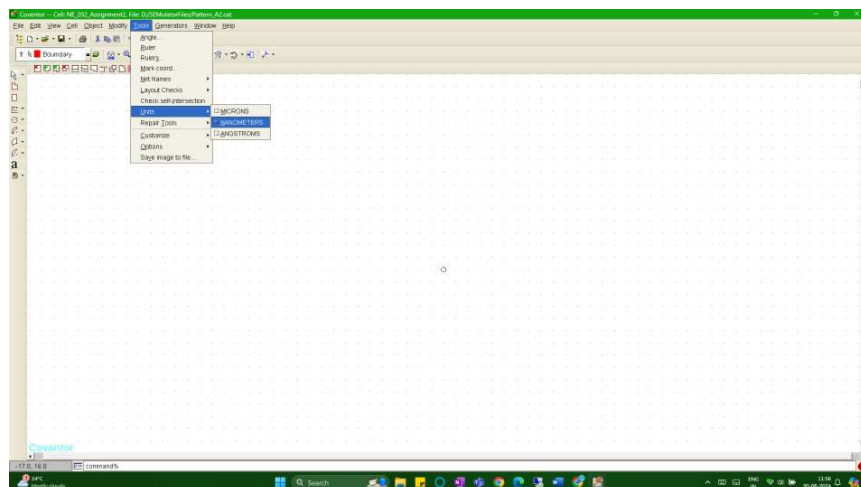


Figure 2 : Setting Unit to nanometres

Step 2: Boundary for the wafer is created using rectangle tab in available shapes in Layout Editor.

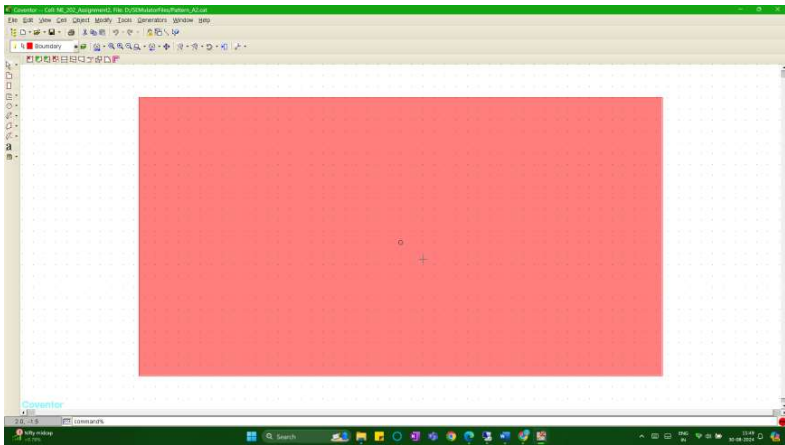


Figure 3 : Boundary of the wafer

Step 3: Rectangular pads have been crated for trenches.

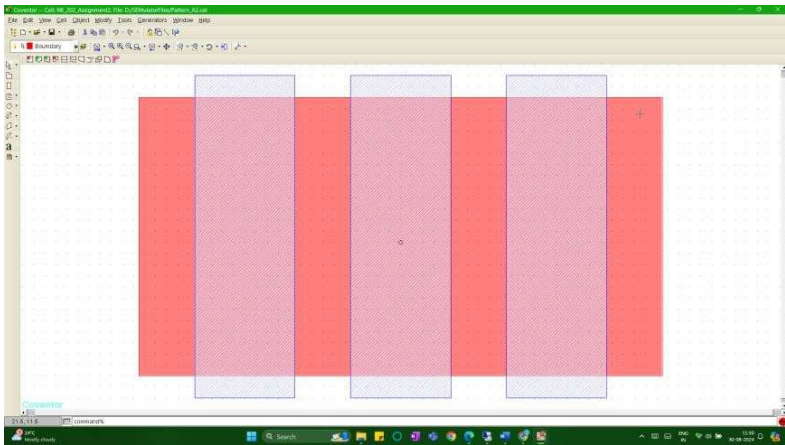


Figure 4 : Mask for patterning trenches

Step 4: Different layers are assigned and beautified.



Figure 5 : Different Layers for Pattern creation

Step 5: Finally, pattern is saved for future uses.

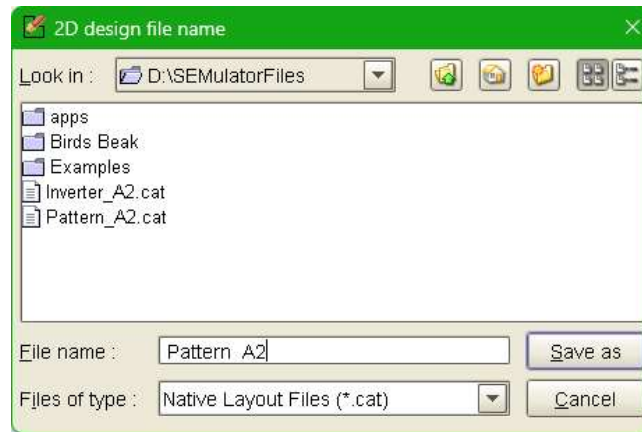


Figure 6 : Saving the patterned Layout

The final pattern looks as shown by Figure 7.

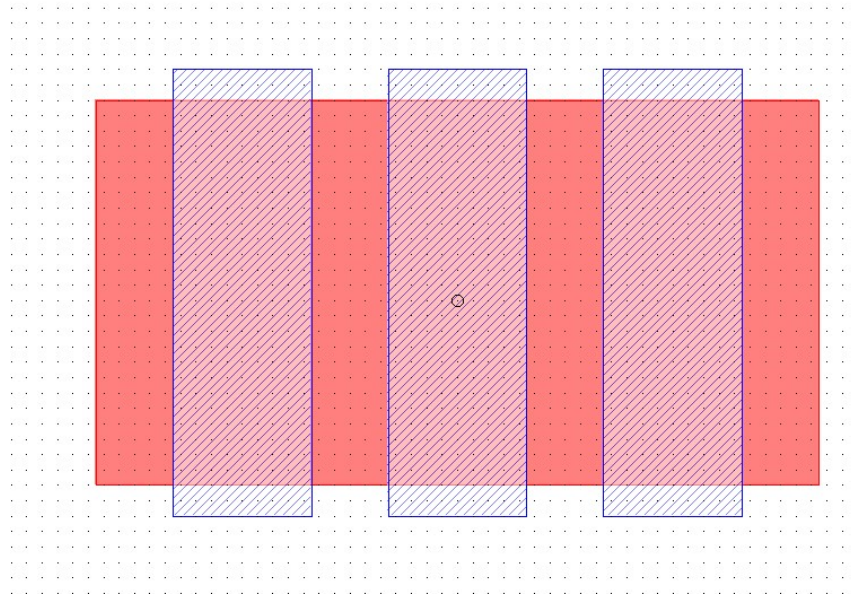


Figure 7 : Final pattern

Question 2

For second task we must make a layout for CMOS inverter circuit cross- section shown by Figure 8 (Left), gate and drain of both PMOS and NMOS are connected to a common gate and drain terminal respectively.

The substrate used is N-type silicon in which p-well process is carried out and source drain of each MOS is made using other type of doping and gate is fabricated using polysilicon. At last, metal contacts is taken out to bias the device.

Figure 8 (Right) shows the circuit diagram of a CMOS inverter, it contains a PMOS and NMOS in series, with shorted gates where input (A) is applied, output (Q) is taken from the connected drains of PMOS and NMOS. V_{dd} and V_{ss} are supplied voltage to provide DC bias to circuit at source of each MOSFET.

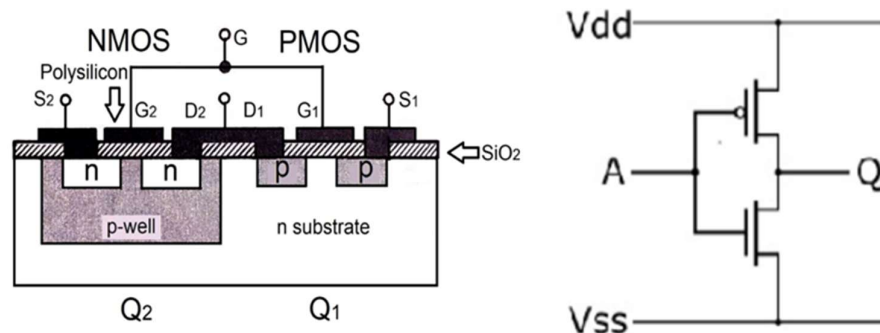


Figure 8 : CMOS Invertor {Left – Cross section; Right – Circuit}

CMOS inverter fabrication steps and layout

First for layout different layers are created as shown by Figure 9, Different layer will be assigned to the shapes created afterwards.

cat:layerbrowser						
File Layer						
↑ ↓ 68 ?						
Layer Name	Color	Fill	V	S	GDS	
N-type-Wafer	red	solid	☒	☒	☒	L0D0
P-type Well	blue	solid	☒	☒	☒	L1D0
NMOS S D	red	left-hatch	☒	☒	☒	L2D0
PMOS S D	blue	left-hatch	☒	☒	☒	L3D0
SiO2 etch	green	solid	☒	☒	☒	L4D0
PolySi S D	black	right-hatch	☒	☒	☒	L5D0
PolySi G D Connection	black	right-hatch	☒	☒	☒	L6D0
Metalization	gray	solid	☒	☒	☒	L7D0

Figure 9 : Different Layers for layout

The steps of the p-well process of CMOS fabrication (Figure 10) are as follows,

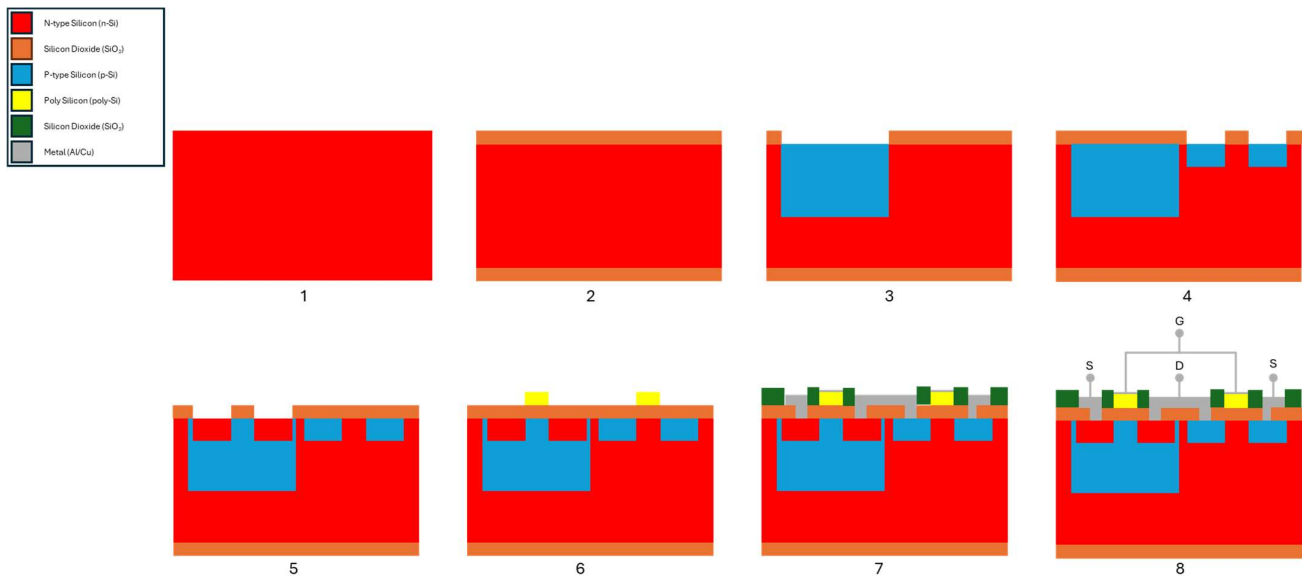


Figure 10 : Process flow for CMOS inverter

Step 1: The process starts with an n-type substrate. A thin SiO_2 layer is deposited which acts as a field oxide for electrical isolation and surface protection which does not require a mask/photolithography step.

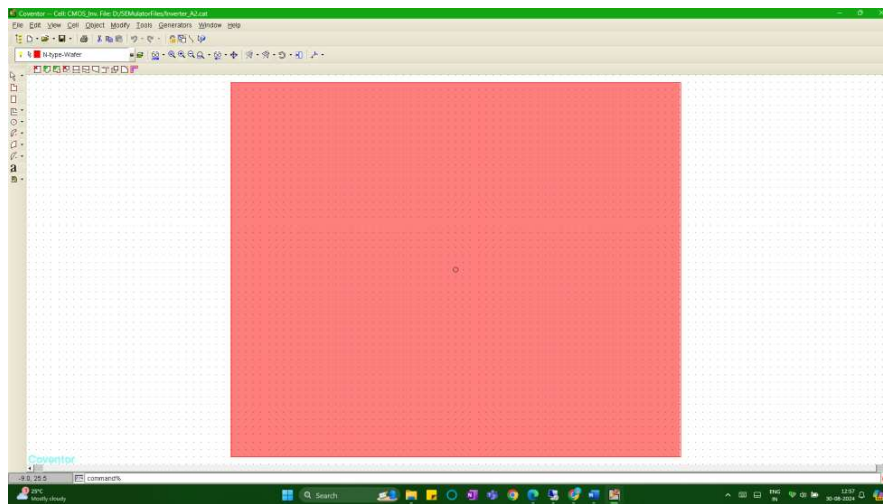


Figure 11 : N-type wafer Boundary

Step 2: Formation of P-well will take place for making NMOS which will require us to remove oxide on the top and implant Boron in substrate, and oxide restored after implantation.

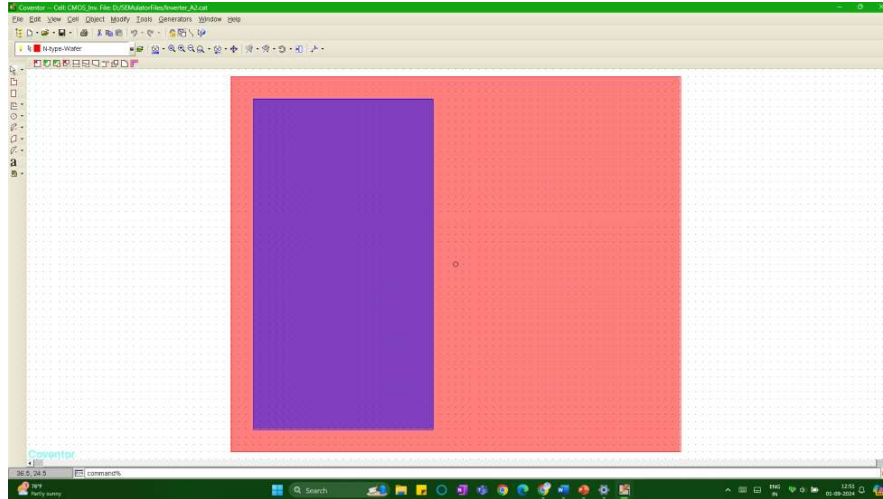


Figure 12 : Formation of P-well

Step 3: Diffusion of Boron is carried out for source and drain regions of PMOS after doing photolithography and etching the SiO₂ layer deposited in window required for diffusion.

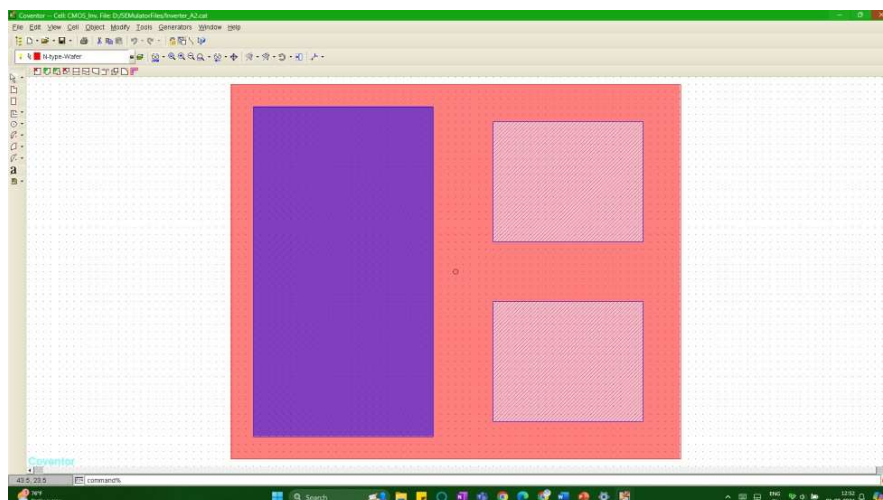


Figure 13 : PMOS Source drain mask

Step 4: Again, we grow SiO_2 , and diffuse Phosphorus for NMOS source and drain by making trenches in SiO_2 using lithography.

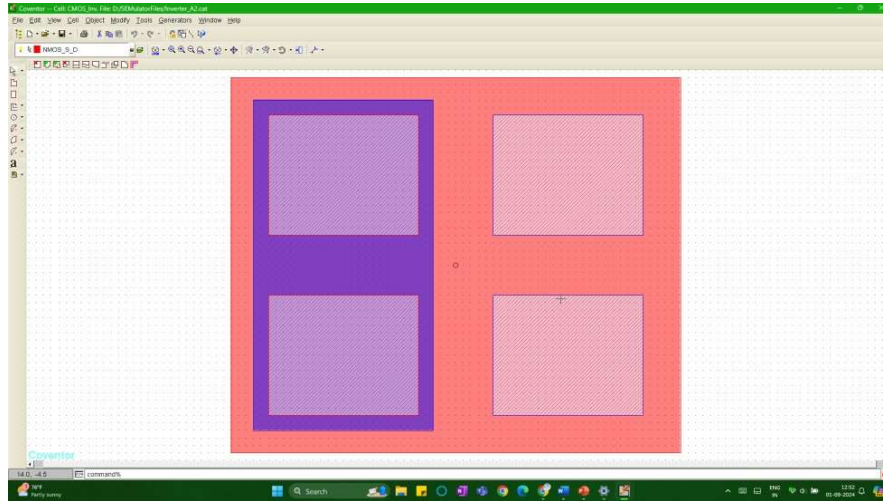


Figure 14 : NMOS Source drain mask

Step 5: After creating source and drains for both NMOS and PMOS we insulate wafer with SiO_2 and make patterns for Poly-silicon deposition on source and drain.

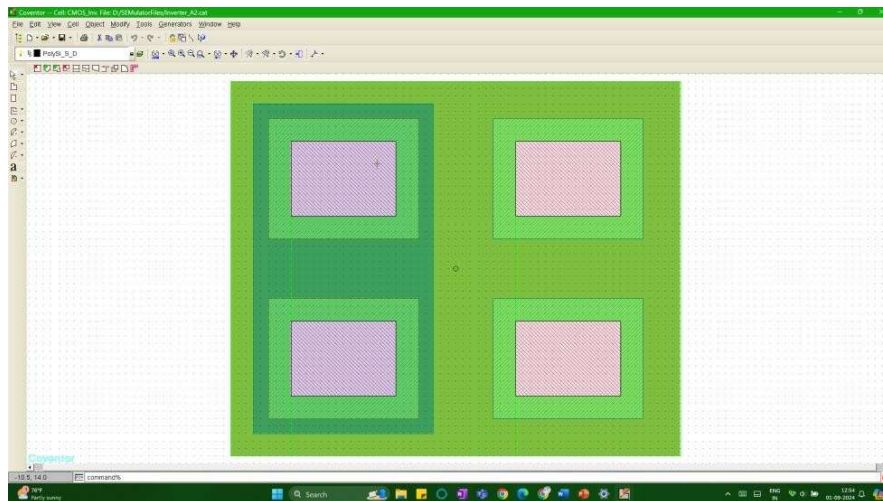


Figure 15 : Polysilicon for source and drain for both MOS

Step 6: After depositing polysilicon for drains and sources we make poly-silicon gate and short drains of both NMOS and PMOS.

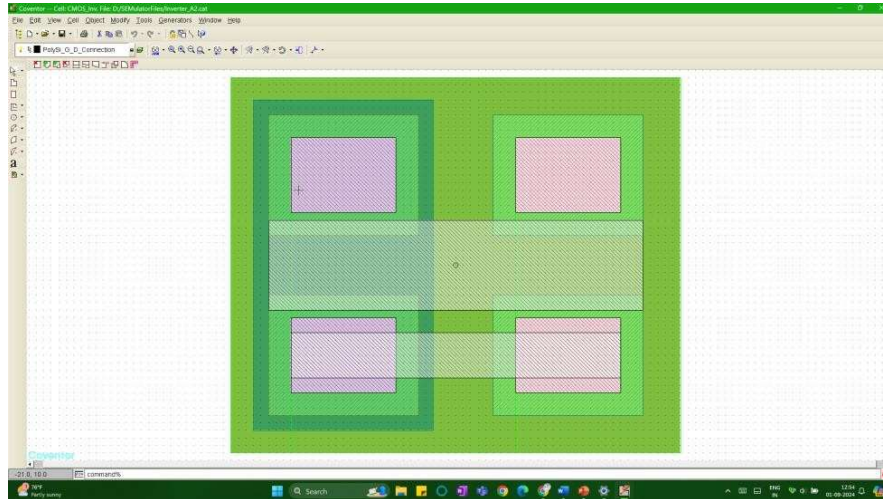


Figure 16 : Shorting gate and drains of both MOS

Step 7: Creation of gate leads us to final step of metallization in which we take metal contacts from source, drain and gate terminal of both NMOS and PMOS.

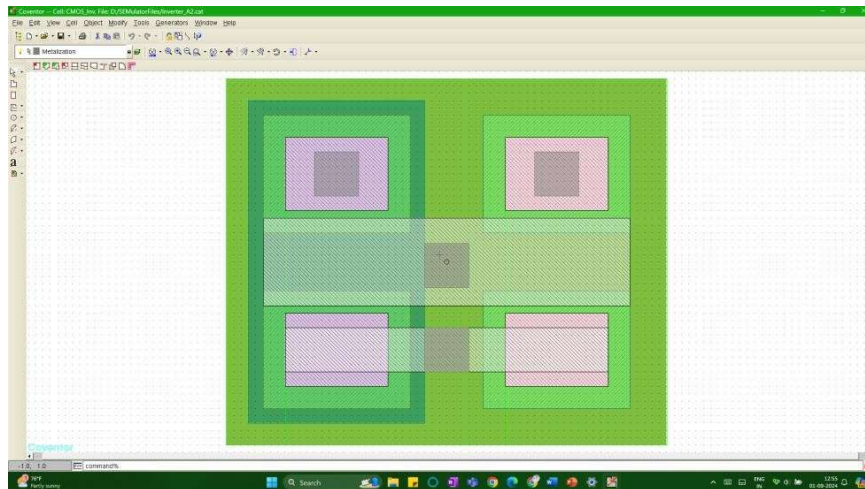


Figure 17 : Final mask